7	**(a)**		When the energy level of atom or system (not electron) at ($n = \infty$) is 0 eV, this is where electron just breaks free from the atom. When the atom is at energy level ($n < \infty$), its electron is bounded to the nucleus by electric attractive force since the nucleus and the electron are of oppositely charged. The atom needs to gain energy to cause electron to break free from it. OR When the atom is at energy level ($n < \infty$), its electron is bounded to the nucleus by electric attractive force since the nucleus and the electron are of oppositely charged. Positive work is done by external agent when the electron breaks free from it. By conservation of energy, energy level of atom at ($n < \infty$) is less than the energy level at ($n = \infty$); i.e. energy level at ($n < \infty$) is of negative value.	B1 B1 B1
	(b)		Transition A.	A1
	(c)		$\Delta E_A = -0.54 - (-13.6) = 13.06$ eV $= 13.06 \times 1.60 \times 10^{-19} = 2.0896 \times 10^{-18}$ J $\lambda_A = \dfrac{6.63 \times 10^{-34} \times 3 \times 10^{8}}{2.0896 \times 10^{-18}} = 9.52 \times 10^{-8}$ m = 95.2 nm (e.c.f.) UV region. (e.c.f. is based on student's calculated wavelength)	M1 A1 A1
	(d)	**(i)**	The incoming electron can give up a fraction of its energy to excite the atom to energy levels ($n = 3$) or ($n = 2$). After which, atom can de-excite from energy levels ($n = 3$) to ($n = 2$), followed by ($n = 2$) to ($n = 1$) , or ($n = 3$) to ($n = 1$) , or ($n = 2$) to ($n = 1$).	B1 A1
		(ii)	**No possible transition.** since photon energy does not correspond **exactly** to the energy interval / difference / gap between ($n>1$) and ($n=1$) .	A1 B1
	(e)		The emission spectrum can be used to <u>determine the composition of a material</u> in atomic emission spectroscopy. It can helps to create special <u>advertisement lightings or entertainment lightings</u>. OR any plausible applications	A1
			Total:	12

8	**(a)**	The **net external force acting on a body is zero**	B1